

Practical no. 6: Design a webpage using html, css and js that prompts user to input a starting number and ending number and display all prime numbers within the specified range.

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Prime Numbers Finder</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      max-width: 600px;
      margin: 20px auto;
      padding: 20px;
      background-color: #f0f0f0;
    }
    .container {
      background-color: white;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
    }
    .form-group {
      margin-bottom: 15px;
    }
    label {
      display: block;
```

```
margin-bottom: 5px;
font-weight: bold;
}
input {
width: 100%;
padding: 8px;
border: 1px solid #ccc;
border-radius: 4px;
box-sizing: border-box;
}
button {
background-color: #4CAF50;
color: white;
padding: 10px 20px;
border: none;
border-radius: 4px;
cursor: pointer;
}
button:hover {
background-color: #45a049;
}
.result {
margin-top: 20px;
font-size: 16px;
color: #333;
}
.error {
color: red;
font-size: 14px;
margin-top: 5px;
}
```

```
#primeList {
  list-style-type: none;
  padding: 0;
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
}

#primeList li {
  background-color: #e8f4f8;
  padding: 5px 10px;
  border-radius: 4px;
}

</style>
</head>
<body>
  <div class="container">
    <h2>Prime Numbers Finder</h2>
    <div class="form-group">
      <label for="startNumber">Starting Number</label>
      <input type="text" id="startNumber" placeholder="Enter starting number">
      <div id="startError" class="error"></div>
    </div>
    <div class="form-group">
      <label for="endNumber">Ending Number</label>
      <input type="text" id="endNumber" placeholder="Enter ending number">
      <div id="endError" class="error"></div>
    </div>
    <button onclick="findPrimeNumbers()">Find Prime Numbers</button>
    <div class="result">
      <h3>Prime Numbers:</h3>
      <ul id="primeList"></ul>
    </div>
  </div>
</body>
</html>
```

</div>

</div>

<script>

```
function isPrime(num) {  
  if (num <= 1) return false;  
  if (num === 2) return true;  
  if (num % 2 === 0) return false;  
  for (let i = 3; i <= Math.sqrt(num); i += 2) {  
    if (num % i === 0) return false;  
  }  
  return true;  
}  
  
function findPrimeNumbers() {  
  const startInput = document.getElementById('startNumber').value.trim();  
  const endInput = document.getElementById('endNumber').value.trim();  
  const startError = document.getElementById('startError');  
  const endError = document.getElementById('endError');  
  const primeList = document.getElementById('primeList');  
  
  // Reset errors and results  
  startError.textContent = "";  
  endError.textContent = "";  
  primeList.innerHTML = "";  
  
  // Validate inputs  
  if (!startInput) {  
    startError.textContent = 'Please enter a starting number';  
    return;  
  }  
}
```

```

    if (!endInput) {
        endError.textContent = 'Please enter an ending number';
        return;
    }

    const start = parseInt(startInput);
    const end = parseInt(endInput);

    if (isNaN(start)) {
        startError.textContent = 'Starting number must be a valid integer';
        return;
    }
    if (isNaN(end)) {
        endError.textContent = 'Ending number must be a valid integer';
        return;
    }
    if (start > end) {
        startError.textContent = 'Starting number must be less than or equal to ending number';
        return;
    }
    if (start < 0) {
        startError.textContent = 'Starting number cannot be negative';
        return;
    }

    // Find prime numbers
    let primes = [];
    for (let num = start; num <= end; num++) {
        if (isPrime(num)) {
            primes.push(num);
        }
    }

```

```
}

// Display results
if (primes.length === 0) {
    primeList.innerHTML = '<li>No prime numbers found in the specified range</li>';
} else {
    primes.forEach(prime => {
        const li = document.createElement('li');
        li.textContent = prime;
        primeList.appendChild(li);
    });
}
}
</script>
</body>
</html>
```

## Prime Numbers Finder

Starting Number

Ending Number

Find Prime Numbers

Prime Numbers:

5 7 11 13 17 19 23 29 31 37 41 43  
47